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Level 1 — 100 marks/question

L1-Q1. (E) Perfect Number

P

CODE

```
n = int(input("Given Number: "))
sum = 0
for i in range(1, n):
    if n % i == 0:
        sum += i
if sum == n:
    print("Its a Perfect Number")
else:
    print("Its not a Perfect Number")
```

INPUT

```
6
```

OUTPUT

```
Given Number: Its a Perfect Number
```

L1-Q2. (E) Min and Max

P

CODE

```
import re

arr_input = input()
numbers = list(map(int, re.findall(r'-?\d+', arr_input)))

m_input = input()
M = int(re.findall(r'-?\d+', m_input)[0])

n_input = input()
N = int(re.findall(r'-?\d+', n_input)[0])

if M == 0 or M > len(numbers) or N > len(numbers):
    print("Invalid Input")
else:
    sorted_desc = sorted(numbers, reverse=True)
    sorted_asc = sorted(numbers)

    mth_max = sorted_desc[M - 1]
    nth_min = sorted_asc[N - 1]

    suffix = lambda n: "st" if n == 1 else "nd" if n == 2 else "rd" if n == 3 else "th"

    print(f"{M} {suffix(M)} Maximum Number = {mth_max}")
    print(f"{N} {suffix(N)} Minimum Number = {nth_min}")
    print(f"Sum = {mth_max + nth_min}")
    print(f"Difference = {abs(mth_max - nth_min)}")
```

INPUT

```
(14, 16, 87, 36, 25, 89, 34)
M = 1
N = 3
```

OUTPUT

```
1 st Maximum Number = 89
3 rd Minimum Number = 25
Sum = 114
Difference = 64
```

L1-Q3. (E) Upper-Lower

P

CODE

```
sentence = input().strip()

upper_sentence = sentence.upper()
print(upper_sentence)

words = sentence.split()
count_a = sum(1 for w in words if w and w[0].lower() == 'a')

print("Total number of words starting with letter A' =", count_a)
```

INPUT

```
advancement and application of information technology are ever changing
```

OUTPUT

```
ADVANCEMENT AND APPLICATION OF INFORMATION TECHNOLOGY ARE EVER CHANGING
Total number of words starting with letter A' = 4
```

L1-Q4. (E) RT-Pattern

P

CODE

```
s = input().strip()
ch = s[-1]

s2 = input().strip()
n = int(s2.split(':')[-1])

for i in range(1, n + 1):
    line = (ch + " ") * i
    print(line.rstrip())
```

INPUT

```
Enter the Character to be printed:+
Number of rows.:5
```

OUTPUT

```
+
+ +
+ + +
+ + + +
+ + + + +
```